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MLGCN-Driver: a cancer driver gene identification method based on multi-layer graph convolutional neural network

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Abstract

Background The progression of cancer is driven by the accumulation of mutations in driver genes. Many researches promote to identify cancer driver genes. However, most of them ignore the high-order features in the network.

Result In this study, we propose a novel method MLGCN-Driver based on multi-layer graph convolutional neural networks (GCN) to boost driver gene identification. MLGCN-Driver employs multi-layer GCN with initial residual connections and identity mappings to learn biological multi-omics features within biological networks. In addition, node2vec algorithm is used to extract the topological structure features of the biological network, and then the features are fed into another multi-layer GCN for feature learning. Meanwhile, the initial residual connections and identity mappings mitigate the over-smooth of features. Finally, the probability of each gene being a driver gene is calculated based on low-dimensional biological features and topological features.

Conclusion We applied the MLGCN-Driver on pan-cancer dataset and cancer type-specific datasets. Experimental results demonstrate the excellent performance of MLGCN-Driver in terms of the area under the ROC curve (AUC) and the area under the precision-recall curve (AUPRC) when compared with state-of-the-art approaches.

Keywords Cancer driver gene, Multi-layer graph convolutional neural networks, Multi-omics feature

Background

It is widely acknowledged that cancer is induced by the accumulation of genetic mutations. Only a small portion of these mutations will confer a selective growth advantage to the mutated cells. Mutations which are essential for cancer development are called driver mutations [1, 2]. The genes with driver mutations are called driver genes. Accurately identifying cancer driver genes is crucial for advancing cancer diagnosis and treatment. It is also one purpose of cancer genomic research [3–5]. Advances in biotechnology and the implementation of projects such as The Caner Genome Atlas (TCGA) [6], the International Cancer Genome Consortium (ICGC) [7] and the Catalogue Of

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Somatic Mutations In Cancer (COSMIC) [8], offer us numerous multi-omics data, which facilitate the generation of many computational methods for identifying cancer driver genes.

Existing methods for identifying cancer driver genes include gene mutation frequency-based methods, network-based methods and machine learning-based methods. Gene mutation frequency-based methods, such as MuSic [9], MutSigCV [10], OncodriveCLUST [11], and ActiveDriver [12], generally identify the genes with higher mutation frequencies than the predefined background mutation rate as cancer driver genes. However, it's hard to estimate the background mutation rate. And these methods also pose challenges in identifying cancer driver genes with low mutation frequency. Some network-based methods assume that cancer driver genes work together in some pathways and protein complexes, rather than a single gene [13]. For example, HotNet2 [14] and MUFFINN [15] consider that the cancer driver genes are frequently enriched within protein complexes or pathways. Some network-based methods use network diffusion or network control theory-based methods [16–18] to identify cancer driver genes. However, the unreliability of the protein-protein interaction (PPI) network limits the efficacy of these network-based methods [19]. To enhance PPI reliability and fully leverage valuable multi-omics data, various approaches [20–24] have been developed. These approaches utilize multi-omics data to filter out noisy connections under the constraints of inter-molecular co-expression, co-function, co-subcellularity, and co-tissue. Nevertheless, these methods do not effectively leverage the relationships within multi-omics data to enhance the accuracy of driver gene prediction.

With the rapid advancement in machine learning (ML) technologies, a myriad of ML-driven approaches have demonstrated noteworthy proficiency in identifying cancer driver genes [25]. Methods such as TUSON [26], LOTUS [27], 20/20+ [28] and DriverML [29] often prioritize the effective features as the inputs of classifier. Subsequently, they leverage techniques like LASSO regression, support vector machines (SVMs), and random forest models to train and predict cancer driver genes, respectively. This paradigm underscores the potential of ML in unraveling the complex landscape of cancer genomics. While traditional machine learning-based methods have achieved satisfactory results in identifying cancer driver genes, they solely rely on genomic multi-omics features, which are insufficient to capture the domain knowledge relevant to cancer.

Recently, deep learning methods have achieved remarkable progress in the fields of molecular biology and genomics. Specifically, several deep learning-based methods for cancer driver gene identification have emerged, particularly focusing on graph neural networks (GNNs). GNN is a novel network representation method that can learn low-dimensional node representation through network structure and node features [30]. For example, EMOGI [31] is a GCN-driven approach that incorporates genomic, epigenomic, and transcriptomic data as gene features. It leverages the PPI network to capture more intricate gene characteristics and show superior accuracy in predicting driver genes across diverse cancers. MTGCN [32] improves predictive accuracy by introducing multi-task learning and network topological features derived from DeepWalk [33]. Specifically, it integrates biological features with network topological features to form enhanced features and adopts an auxiliary task of link prediction to guide model training based on Chebyshev convolution network. Some methods use graph diffusion to identify cancer driver genes [34, 35]. Especially, HGDC [36] utilizes personalized PageRank

(PPR) method [37] to generate auxiliary networks on the original biological network and designs a novel message aggregation scheme for GCN to identify driver genes. PDGCN employs GCN integrated with Conditional Random Fields (CRF) to predict cancer driver genes at the individual level [38]. SSCI uses a positive-unlabel (PU) learning algorithm to infer reliable negative samples and performs feature masking on the parameterized network, while using GCN for feature denoising [39]. MHN-GCN are based on four omics data and user GAT and GATv2 to enhance the ability to predict cancer driver genes [40]. MNGCL constructs multiple biological networks and employs GCL (Graph Contrastive Learning) to learn feature representations of gene [41]. SGCD employs GCN with representation separation to replace the traditional message-passing mechanism, and designs a bimodal feature extractor to separately capture the topological information of the graph and the multi-omics feature information [42]. These methods have effectively improved driver gene prediction accuracy. However, biological networks are inherently sparse, and the aforementioned GCN-based methods normally utilize shallow GCN architectures, which limit the ability of the model to capture the features of high-order neighbors in the network. In addition, during the information aggregation process of GCN, the distinct features of driver genes may be easily smoothed by the features of their neighboring non-driver genes.

To address the aforementioned issues, we propose MLGCN-Driver, a novel framework that enhances cancer driver gene identification based on biomolecular network through multi-layer GCN. Firstly, we employ multi-layer GCN to learn information of high-order neighbors of nodes for biological networks with multi-omics features and topological features respectively. This allows us to capture information between genes of the same type that are far away from each other. Secondly, we integrate initial residual connections and identity mappings in each layer of GCN. The main process of initial residual connection is to add some original features to each layer of GCN after learning. The identity mapping is to add an equally identity matrix to the weight matrix in each layer of GCN. These two strategies are incorporated in GCN to prevent the unique features of driver genes from being smoothed by the features of neighboring non-driver genes. Furthermore, node2vec [43], a random walk algorithm with both depth-first and breadth-first search, is used to extract topological features of the network. Finally, the prediction from two multi-layer GCN predictors are fused for the prediction of cancer driver genes.

Methods

Overview of MLGCN-Driver framework

MLGCN-Driver primarily consists of three parts, as illustrated in Fig. 1. Module (a) is data collection and preprocessing step. In this step, we collect multi-omics data containing somatic mutation, gene expression, and DNA methylation. These three types of multi-omics data are concatenated with system-level data to form the biological features of each gene. Module (b) is gene representation learning step. In this step, node2vec algorithm is used to extract network topological features from PPI network. Consequently, to learn useful higher-order neighbor information in the network, the obtained biological features and network topological features are sent to multi-layer GCN separately. Module (c) is driver gene prediction step. Two types of learned features are used to predict the importance of genes individually. To obtain accurate and reliable prediction results, MLGCN-Driver applies a weighted approach to fuse the outputs of the two

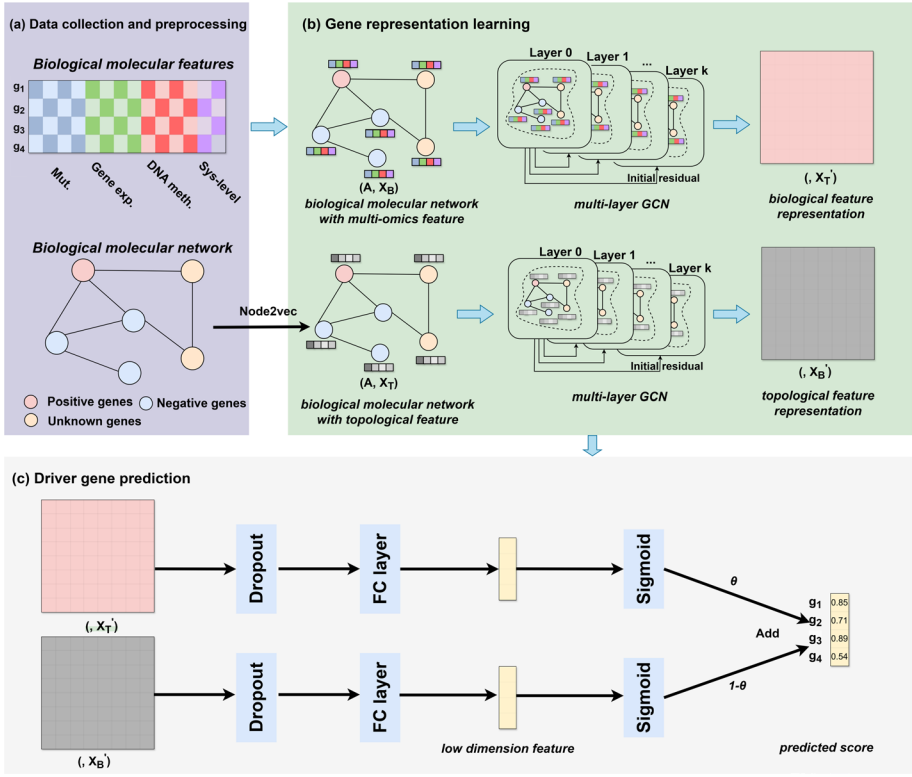

Fig. 1 The overall workflow of MLGCN-Driver. **a** Data collection and preprocessing: A subset of genes networks and representations are obtained from databases. **b** Gene representation learning: Topological features are obtained from node2vec algorithm and sent with biological features into multi-layer GCN for feature learning. **c** Driver gene prediction: After feature learning, the high dimensional features are transformed into low-dimensional features through dropout layer and fully connected layer. Then, the predicted scores for each gene are obtained through the sigmoid layer. Mut: somatic mutation, Gene exp.: gene expression, DNA meth.: DNA methylation and Sys-level: system-level feature

predictors for cancer driver gene prediction. This methodology enables us to fully leverage the advantages of both biological features and network topological features, thereby enhancing the accuracy and stability of our predictions.

Data collection and preprocessing

The datasets used in MLGCN-Driver include multi-omics data from pan-cancer and type-specific cancer, as well as biomolecular networks. These data are derived from the HGDC [36] and MTGCN [32] studies. In the following section, we will elaborate on the data in detail.

In the pan-cancer dataset, each gene is represented by 58-dimensional multi-omics features, including 48-dimensional molecular features and 10-dimensional system-level features. The 48-dimensional molecular features are somatic mutation, DNA methylation, and gene expression which are collected for 16 different cancer types in the TCGA database. In detail, mutation, DNA methylation, and gene expression data for 29,446 patients with 16 cancer types (i.e., BLCA, BRCA, CESC, COAD, ESCA, HNSC, KIRC, KIRP, LIHC, LUAD, LUSC, PRAD, READ, STAD, THCA, UCEC) are from the TCGA database. For each cancer type, the mutation frequency is calculated by dividing the number of non-silent SNVs by the exon gene length. The differential DNA methylation is quantified through the average of the methylation data signals between tumor and

normal samples across samples. The differential expression level of a gene is measured by log2 fold change in tumor expression relative to normal samples and then averaging over all samples for each cancer type. The system-level features are from sysSVM2 [44] tool and capture global gene characteristics which do not directly originate from cancer type-specific omics data, including gene duplication, gene essentiality, tissues expressing gene, topological properties of biomolecular network, etc.

To comprehensively evaluate the performance of MLGCN-Driver, we conducted experiments on three biomolecular networks for the pan-cancer dataset. In detail, they are the pathway network comprising KEGG and Reactome pathways (PathNet [45]), the gene-gene interaction network extracted from the Encyclopedia of RNA Interactomes (GGNet [46]), and the protein-protein interaction network collected from the STRING database (PPNet [47]). Specifically, PathNet has 7,699 nodes and 92,710 edges, GGNet has 11,309 nodes and 621,988 edges, and PPNet has 11,473 nodes and 285,843 edges.

In this study, to validate the performance of MLGCN-Driver in specific cancer, we apply MLGCN-Driver to four kinds of cancer, Breast cancer (BRCA), Lung adenocarcinoma (LUAD), Bladder cancer (BLCA) and Hepatocellular carcinoma (LIHC). For each type of cancer, the somatic mutation, DNA methylation, and gene expression are extracted from pan-cancer dataset, constituting 3-dimensional biological features and 10-dimensional system-level features. The biological network used is PathNet mentioned in the above pan-cancer dataset. In order to maintain consistency with the dimensionality of biological features, the network topological features' dimension is also 13.

The driver genes (positive samples) and non-driver genes (negative samples) for the pan-cancer dataset are the same as HGDC [36]. The positive samples include high-confidence cancer driver genes collected from the NCG [48] database and excavated from the literature, and the negative samples are the remaining genes after removing genes appearing in the NCG, COSMIC, OMIM, and KEGG databases from all the collected genes. Therefore, the pan-cancer dataset consists of 796 positive samples and 2,187 negative samples.

For the can cer type-specific datasets, the positive samples are driver genes for specific cancer in NCG database, including four cancer type-specific datasets: BRCA with 202 positive samples, LUAD with 179 positive samples, BLCA with 95 positive samples, and LIHC with 82 positive samples. The negative samples in the cancer type-specific datasets are the same as those in the pan-cancer dataset, totaling 2,187 negative samples.

Gene representations learning

Topological feature extraction

Node2vec is a widely used algorithm for network representation learning, which captures both local and global topological features through biased random walks. It has been successfully applied in several previous studies to extract meaningful topological representations from biological networks [32]. Motivated by its proven effectiveness, we also adopt node2vec in this study. Its core concept is to learn node representations through random walks. The algorithm balances local and global network information by adjusting two parameters during the random walk: the return parameter p and the in-out parameter q. When p and q are small, the random walk is more likely to return to the previously visited node while exploring regions that are farther away, which aids in capturing the structural characteristics of the network. In this study, we set both p and

q to 0.5, encouraging the random walk to capture both local structures and broader network patterns. The influence of parameters p and q on model performance is presented in Table S5 of the Supplementary file. Specifically, for the pan-cancer dataset, the edge set E of the biological network is fed into the node2vec algorithm, resulting in 58-dimensional network topological features, denoted as X_T.

Feature learning with multi-layer GCN

GCN is a deep learning model specifically designed to handle graph-structured data, such as social networks, molecular structures, transportation networks, etc. The core idea of GCN is to extend the convolutional operation from traditional grid data structures to graph data structures. In GCN, the convolutional operation is redefined as a process of passing and aggregating information between nodes and their neighbors within the graph. Through this approach, GCN is able to capture the topological structure of the graph and the complex relationships among nodes.

Specifically, given graph G with adjacency matrix A, the graph convolution operation on graph G can be defined as:

$$H^{(l+1)} = \sigma(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(l)}W^{(l)}) \tag{1}$$

where $\tilde{D} = D + I$, $\tilde{A} = A + I$. D is the degree matrix, I is the identity matrix, A is the adjacency matrix, l is the number of layers, $H^{(l)}$ is the node feature representations of the l-th layer GCN, $W^{(l)}$ is the learnable weight matrix for the l-th layer GCN and $\sigma(\cdot)$ denotes a nonlinear activation function ReLU.

In this study, MLGCN-Driver employs multi-layer GCN to learn feature representations of genes, as illustrated in Fig. 1b. Unlike traditional training methods, we first use two independent multi-layer GCNs to train biological features separately from network topological features. By propagating information through layers, it can capture features from high-order neighbors, resulting in richer node representations. Additionally, as the number of layers increases, GCN can gradually learn features from local to global scales in the graph. Traditional two convolutional layers mainly focus on local neighborhood information of nodes, while more layers are able to capture broader graph structural information.

Secondly, while using multi-layer GCN to relieve over-smoothing, initial residual connections and identity mappings which have been validated the effectiveness in GCNII [49] are adopted. On the one hand, the initial residual connections ensure that, even after stacking multiple layers, the final feature representation of each node retains at least α_l times the initial features, where α_l can be simply set to 0.1 or 0.2. On the other hand, identity mappings aim to preserve part of the current layer's GCN features that ensure it is not inferior to shallower models. Identity mappings incorporate a portion (β_l) of the identity matrix I into the weight matrix $W^{(l)}$ during the message aggregation process of each GCN layer. In which the hyperparameter β_l is designed to ensure that as the number of GCN layers increases, the weight matrix decreases accordingly. And β_l is defined as:

$$\beta_l = \log\left(\frac{\lambda}{l} + 1\right) \approx \frac{\lambda}{l} \tag{2}$$

where λ is a hyperparameter, l is the current number of layers in GCN.

After incorporating these two techniques, the formula for the graph convolutional operation performed by the GCN can be formulated as:

$$H^{(l+1)} = \sigma(((1-\alpha_l)\tilde{P}H^{(l)} + \alpha_l H^{(0)})((1-\beta_l)I + \beta_l W^{(l)})) \tag{3}$$

where $\tilde{P} = \tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}$, α_l and β_l are hyperparameters for the initial residual connection and identity mapping, respectively. $H^{(l)}$ represents the node feature representation of the l-th layer GCN, $H^{(0)}$ denotes the original input features, I is the identity matrix, $W^{(l)}$ is the learnable weight matrix in the l-th layer GCN, and $\sigma(\cdot)$ represents a nonlinear activation function ReLU.

In this way, the processed 58-dimensional biological feature X_B and 58-dimensional network topological feature X_T are input into two independent multi-layer GCNs for node feature representation learning. Then the learned biological feature $X_B^{'}$ and network topological feature $X_T^{'}$ are fed into module c in Fig. 1 for prediction.

Driver gene prediction

The predictor mainly consists of a fully connected layer and a Sigmoid activation function. After $X_B^{'}$ and $X_T^{'}$ are fed into the predictor, two corresponding predicted values, $pred_B$ and $pred_T$, are obtained. Subsequently, the final predicted value *pred* is calculated by weighting the two predicted values as follows:

$$pred = \theta * pred_B + (1-\theta) * pred_T \tag{4}$$

where θ is a parameter which is set manually. Furthermore, the loss function used in this paper is the classical binary cross-entropy loss function. Specifically, the cross-entropy loss is calculated separately for the biological features and the network topological features, which can be formulated as:

$$L_B = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(y_i^B) + (1-y_i)\log(1-y_i^B)\right] \tag{5}$$

$$L_T = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(y_i^T) + (1-y_i)\log(1-y_i^T)\right] \tag{6}$$

where y_i represents the true label of gene i, y_i^B and y_i^T represent the predicted labels based on biological features $X_B^{'}$ and network topological features $X_T^{'}$ after Sigmoid transformation respectively, and n represents the number of labeled genes in the training set.

The total training loss is obtained by weighting the above two components as follows:

$$L_{total} = \omega L_B + (1-\omega)\, L_T \tag{7}$$

where ω is a parameter that we set manually.

Evaluation metrics

In this paper, cancer driver gene classification is essentially a binary classification task. Therefore, two commonly used evaluation metrics, the area under the receiver operating characteristic curve (AUROC) and the area under the precision-recall curve (AUPR), are employed. The definition of AUROC is as follows

$$AUROC = \frac{\sum_{i=1}^{m-1}(TPR_i + TPR_{i+1}) \times (FPR_{i+1} - FPR_i)}{2} \tag{8}$$

where m represents the number of points on the AUROC curve. *TPR* stands for True Positive Rate and *FPR* stands for False Positive Rate. The definition of AUPR is as follows

$$AUPR = \sum_i p(i)\Delta r(i) \tag{9}$$

where *r(i)* represents the recall rate at the i-th threshold, which is the proportion of true positive samples among the correctly predicted outcomes, and $p(i)$ represents the precision at recall rate *r(i)*.

Results

Comparison results of MLGCN-Driver with other methods on pan-cancer dataset

To evaluate the performance of MLGCN-Driver in identifying cancer driver genes, we compared it with seven other state-of-the-art methods on three networks: GGNet, PathNet, and PPNet. The hyperparameters of MLGCN-Driver vary across different biological networks. This is because different biological networks possess distinct topological structures, including the number of nodes, edge density, node connection patterns, and so on. These structural differences can impact the feature learning performance of the GCN. Moreover, the node features in different biological networks may also vary, and the dimensionality and distribution of these features can affect the learning effectiveness of GCN. The hyperparameters of MLGCN-Driver are detailed in Supplementary materials Table S1. Specifically, the number of GCN layers used for GGNet, PathNet, and PPNet is 8, 4, and 8, respectively. These methods include three advanced GCN-based methods specifically designed for cancer driver gene identification (i.e., HGDC, MTGCN, and EMOGI), three classical GNN methods (i.e., GCN [50], GAT [51], and Chebnet [52]), and one machine learning method (SVM). The comparative methods and their experimental results are derived from [36]. Experiments of our method are conducted on the Ubuntu server with an AMD EPYC 7320P 16-Core Processor (256G RAM) and an Nvidia RTX 3090 GPU (24G GPU RAM). The computational time of different methods on the three networks are provided in Table S6 of the Supplementary file. The AUROC and AUPR of each method under ten times five-fold cross-validation are shown in Fig. 2. And the standard deviations of the experimental results are shown in Table S3 of the Supplementary file.

The results in Fig. 2 indicate that MLGCN-Driver achieves higher AUROC and AUPR values on GGNet, PathNet, and PPNet compared to the other seven methods,

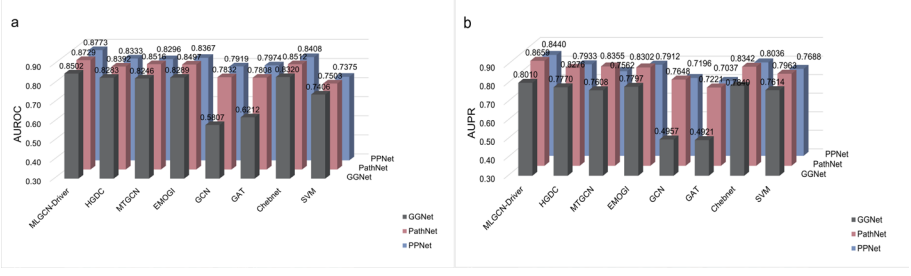

Fig. 2 Results of our MLGCN-Driver with other methods on pan-cancer dataset. **a** Results of AUROC. **b** Results of AUPR

demonstrating the validity of our approach. For example, experimental results on these three different networks demonstrate that MLGCN-Driver consistently outperforms the current state-of-the-art method, with gains of 1.82–3.65% in AUC and 1.6–4.04% in AUPR. Meanwhile, the results show that MTGCN, EMOGI, and Chebnet, which are based on ChebConv, also achieve relatively high AUROC and AUPR values. Because ChebConv uses different weight matrices for feature aggregation, that is, it separates the node's ego-features and neighboring features during the message aggregation. HGDC designs an improved message aggregation and propagation method to facilitate gene representations learning on both original biological network and auxiliary network, which can separate ego-features of genes from neighboring features as well. MLGCN-Driver uses initial residual connections and identity mappings for feature aggregation, which not only considers the gene's own features and its neighboring features but also mitigates over-smoothing due to stacking more GCN layers. At the same time, these two strategies also prevent driver genes' ego-features from being smoothed by surrounding non-driver gene features. The inclusion of these modules enables MLGCN-Driver to achieve optimal results on GGNet, PathNet, and PPNet.

Ablation experiments of MLGCN-Driver

MLGCN-Driver primarily improves the prediction accuracy of the model through the following three aspects: (i) it employs multi-layer GCN to capture the high-order neighbor features of nodes. (ii) While using multi-layer GCN, we incorporate two strategies, initial residual connections and identity mappings, to address the issue of the driver genes' own features being smoothed by the features of surrounding non-driver genes. (iii) Additionally, we utilize the node2vec algorithm to extract topological features of the network, which assists in the prediction of the model. To investigate the contribution of these three modules to MLGCN-Driver, we have designed the following five model variants.

2LGCN: It consists of two GCN layers only based on biological features and its filter size is the same as MLGCN-Driver's.

Multi-layer GCN: It represents adding more layers of GCN based on 2LGCN.

Multi-layer GCN + TF: It represents that we incorporate topological features (TF) extracted by node2vec into the model.

Multi-layer GCN + IRC + IM: It represents incorporating initial residual connections (IRC) and identity mappings (IM) based on Multi-layer GCN.

Multi-layer GCN + IRC + IM + TF: It represents that we add topological features extracted by node2vec into Multi-layer GCN + IRC + IM.

Figure 3 shows a comparison of the results from the ten times five-fold cross-validation conducted on three networks using the pan-cancer dataset for MLGCN-Driver and its variant models. It can be observed that each module has influenced the prediction accuracy of the model on the three networks. 2LGCN serves as the foundation for other models. Its performance in terms of AUROC and AUPR is moderate, indicating room for improvement. Adding additional GCN layers based on the 2LGCN model leads to a decline in the model's performance. This is attributed to the phenomenon of over-smoothing, in which the learned node features become indistinguishable as the layers stack deeper, resulting in a loss of critical information needed for effective feature

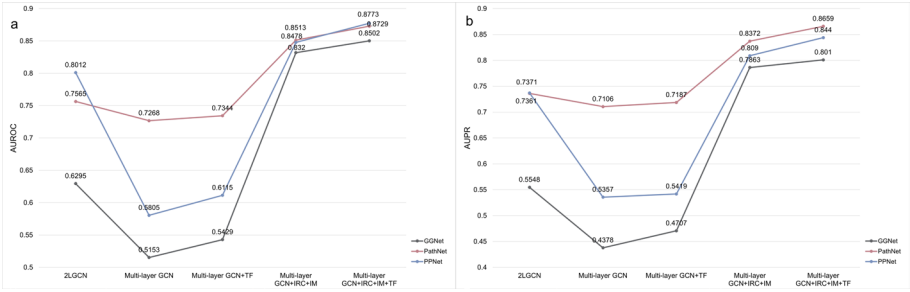

Fig. 3 The results of ablation experiments. **a** Results of AUROC. **b** Results of AUPR

representation. To address this limitation, we incorporate initial residual connections and identity mappings in the subsequent variant of the model. By adding initial residual connections and identity mappings to Multi-layer GCN, the performance of this model on three networks is significantly improved. For example, Multi-layer GCN + IRC + IM improves AUC by 28.91%, 11.69% and 23.63% and AUPR by 31.56%, 11.85% and 26.41% over Multi-layer GCN + TF on GGNet, PathNet and PPNet, respectively. From the results of Multi-layer GCN + TF and Multi-layer GCN + IRC + IM + TF, it can be observed that adding topological features can enhance the model's understanding of the overall network structure, empowering it to more precisely identify cancer driver genes. Finally, MLGCN-Driver employs all the afore-mentioned strategies and achieves the best AUROC and AUPR values among its variants.

Comparison results of MLGCN-Driver with other methods on cancer type-specific datasets

To verify the generalization ability of our model, we also investigated the effectiveness of MLGCN-Driver in predicting driver genes on cancer type-specific datasets. In this study, for ensuring fair and direct comparison, four types of cancer datasets including BRCA, LUAD, BLCA, and LIHC are selected to maintain consistency with previous studies [32]. For each cancer type, we select the corresponding 3-dimensional biological features and 10-dimensional system-level features from the pan-cancer dataset. At the same time, 13-dimensional network topological features are extracted. For comparison, we take the network of PathNet as an example and retrain all the models with cancer type-specific datasets on PathNet. The parameters for the baseline models are configured according to their original setting. The results under ten times five-cross validation shown in Fig. 4. Specifically, the standard deviations of the experimental results are shown in Table S4 of the Supplementary file.

As observed in Fig. 4, MLGCN-Driver exhibits the highest AUROC and AUPR values among the datasets for BRCA, LUAD, BLCA and LIHC. The main reason may be that MLGCN-Driver uses multi-layer GCN, initial residual connections and identity mappings for feature learning, and the features of higher-order neighbors in the network improve model's prediction accuracy. It is evident that MLGCN-Driver has a greater improvement in AUPR. The reason may be that there are few positive samples in the cancer type-specific dataset, and when the model predicts more positive samples, the value of AUPR will be improved. In this study, MLGCN-Driver can not only leverage the advantages of GCN but also incorporate strategies such as initial residual connections and identity mappings, further enhancing the model's prediction accuracy.

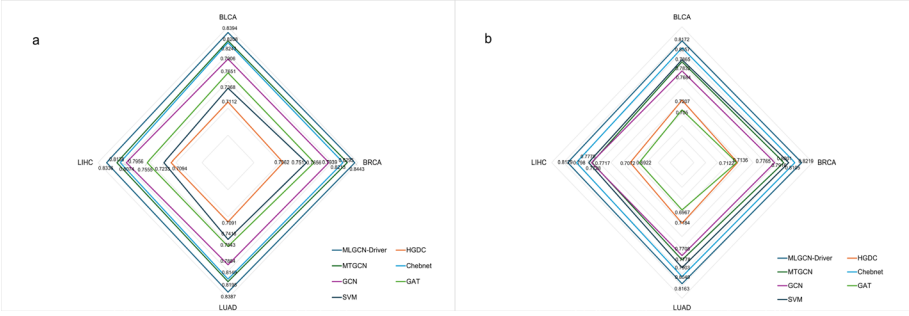

Fig. 4 The results of our MLGCN-Driver with other methods on cancer type-specific datasets. **a** Results of AUROC. **b** Results of AUPR

Table 1 The distribution of the top 100 genes on three networks

Network	Number of true driver genes	Number of candidate driver genes	Number of non-driver genes
GGNet	73	22	5
PathNet	82	11	7
PPNet	79	15	6

Performance of MLGCN-Driver on different biomolecular networks

To investigate the effectiveness of MLGCN-Driver in identifying novel cancer driver genes on different biological networks, we trained the MLGCN-Driver model 100 times on GGNet, PathNet, and PPNet network. For each node in a network, we used the average predicted value over 100 training iterations as the prediction score for subsequent analysis.

Top 100 predicted genes from GGNet, PathNet and PPNet are selected for analysis. By comparing the predicted results with the labeled driver genes in the NCG database (shown in Table 1), we found that 70%-80% of the predicted genes were true driver genes, 10%-20% were potential driver genes, and 10% were non-driver genes. This indicates that MLGCN-Driver is capable of predicting a significant proportion of driver genes accurately.

Next, to study the influence of different biological networks on MLGCN-Driver, we used a Venn diagram to represent the overlap of predicted driver genes in the three networks as well as the true driver genes. As shown in Fig. 5, a certain number of driver genes are predicted only in a specific biological network, indicating that different biological networks indeed have an impact on the model's prediction of driver genes. Furthermore, we found that the proportion of true driver genes predicted in at least two networks is higher than that predicted in a single network. For example, the intersection of predicted driver genes in GGNet, PathNet, and PPNet comprises 44 genes, of which 43 are true driver genes confirmed in the database. However, in GGNet, only 13 out of 30 genes are true driver genes. These results suggest that the intersection of predicted driver genes from multiple networks may yield more reliable results compared to predictions from a single network.

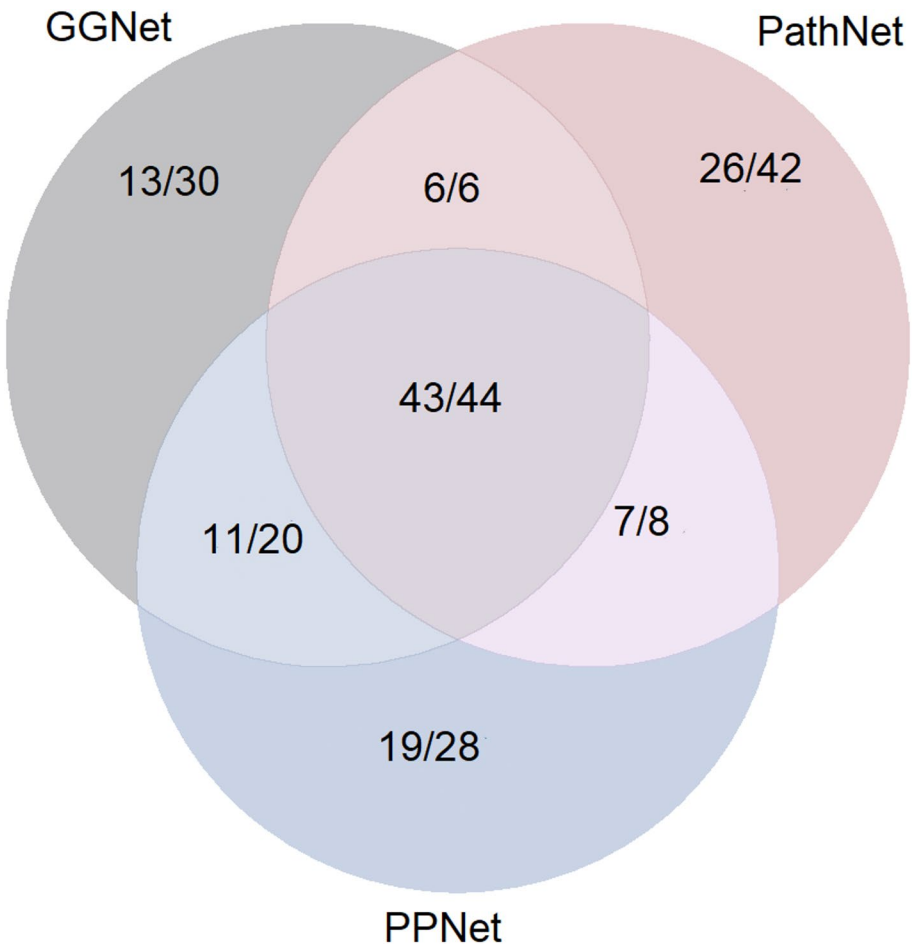

Fig. 5 The overlapping number of driver genes predicted by the three networks. The denominator represents a portion of the top 100 genes predicted by the model, the numerator represents the number of true driver genes

Performance of MLGCN-Driver on novel cancer driver genes

To evaluate the ability of MLGCN-Driver to predict new driver genes, we conducted literature validation on novel genes that do not appear in the NCG database among the top 100 predicted genes, that is, the non-driver genes in Table 1.

Taking novel driver genes in GGNet as an example, we found that these genes with high prediction scores but not listed in the NCG database have literature indicating their key roles in certain cancers (the analysis of novel cancer driver genes on PathNet and GGNet are shown in Supplementary file Text S1). For instance, *ACACA* [53] is considered as a key factor in the prognosis of hepatocellular carcinoma (HCC). Bioinformatics analyses indicated a correlation between elevated *ACACA* protein/mRNA expression and adverse prognosis in HCC patients. Targeted knockdown of *ACACA* significantly impeded HCC cell proliferation, colony formation, migration, invasion, EMT progression, and induced cell cycle arrest.*AGO1* [54] has been shown to potentiate the recruitment of RNA polymerase II to the promoter regions of genes pertinent to cellular growth and survival, with a pronounced effect observed in prostate cancer cells. Furthermore, *AGO1* facilitated transcriptional activation of cancer-associated genes, which occurs in a manner contingent upon the presence of AT repeats within the promoter sequence, has been implicated in the neoplastic progression of malignancies originating

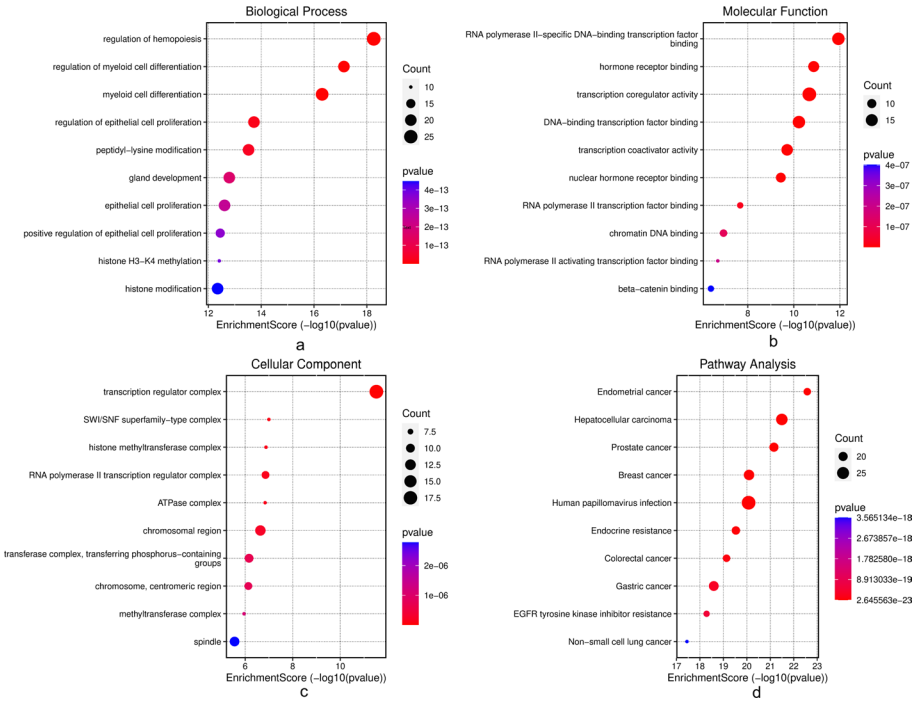

Fig. 6 The GO and KEGG enrichment analysis results of the top 100 driver genes predicted on the GGNet. **a** Results of GO enrichment analysis for biological process. **b** Results of GO enrichment analysis for molecular function. **c** Results of GO enrichment analysis for cellular component. **d** Results of KEGG enrichment for pathway analysis

from diverse tissue types. *FN1* [55] exhibits a predominant expression pattern in cancerous tissues, contrasting with its low or absent expression in adjacent non-tumorous tissues. Within tumor tissues, *FN1* is primarily localized to the stromal compartment. Notably, the elevated expression of extracellular *FN1* has been correlated with a reduction in overall survival (OS), whereas the expression of soluble *FN1* has not demonstrated a similar prognostic association. *TLN1* [56] is a potential therapeutic target for TNBC (Triple Negative Breast Cancer). Notably, functional ablation of both *KIF14* and *TLN1* significantly potentiated chemosensitivity in four TNBC cell lines (MDA-MB-231, HCC38, HCC1937, and Hs478T), whereas this enhancement was absent in three hormone receptor-positive breast cancer cell lines (MCF7, T47D, and HCC1428) as well as normal human mammary epithelial cells (HMEC). Furthermore, heightened expressions of *KIF14* and *TLN1* have been observed in TNBCs compared to other clinical subtypes of breast cancer, underscoring their potential significance in this aggressive subtype.

GO and KEGG pathway enrichment analysis

To gain a better understanding of the function of the genes predicted by MLGCN-Driver, we conducted GO and KEGG enrichment analysis on the top 100 genes predicted across three biological networks. GO enrichment analysis includes biological process, molecular function, and cellular component. KEGG enrichment analysis includes pathway analysis.

Here, we take the results from GGNet as an example, as shown in Fig. 6 (the GO and KEGG pathway enrichment analysis on PathNet and PPNet are shown in Supplementary file Text S2). For GO analysis, in terms of biological process (Fig. 6a), these top 100 genes play significant roles in the regulation of hemopoiesis, regulation of myeloid cell

differentiation, etc. In hematological malignancies, malignant cells remodel their niche into a permissive environment to enhance the survival of leukemic cells. These events are accompanied by loss of normal hematopoiesis [57]. Regulation of myeloid cell differentiation is well established that tumor growth is associated with accumulation of immature myeloid cells (ImC). They play an important role in tumor-associated immune suppression [58]. In terms of molecular functions (Fig. 6b), these top 100 genes possess key functions such as DNA-binding transcription factor binding, transcription co-activator activity. Lambert et al. [59] discusses various ways to target transcription factors in cancer models: by modulating their expression or degradation, by blocking protein/protein interactions, by targeting the transcription factor itself to prevent its DNA binding either through a binding pocket or at the DNA-interacting site, some of these inhibitors being currently used or evaluated for cancer treatment. For transcription co-activator activity, AIB1(amplified in breast cancer 1) is a transcriptional co-activator that promotes the transcriptional activity of multiple nuclear receptors and a number of other transcription factors. At the same time, AIB1 is a member of the p160 nuclear receptor co-activator family and is considered as an important oncogene in breast cancer [60]. In terms of cellular components (Fig. 6c), these top 100 genes are significantly enriched in transcription regulatory complexes and chromatin regions. It has been reported that the paralogous zinc-finger proteins – CTCF and brother of the regulator of imprinted sites (BORIS), directly contribute to transcriptional regulation of NY-ESO-1 in lung cancer cells [61]. In addition, it has been described recent insights into altered genome architecture in human cancer, highlighting multiple pathways toward disruptions of chromatin structure, including structural variation, noncoding mutations, metabolism, and de novo mutations to architectural regulators themselves [62].

The enriched KEGG pathways (Fig. 6d) are related to many kinds of cancers like endometrial cancer, prostate cancer, breast cancer, colorectal cancer, gastric cancer and non-small cell lung cancer. It indicates that MLGCN-Driver's prediction results have certain biological significance in cancer and can effectively predict cancer-related genes.

Based on the results of GO enrichment analysis and KEGG pathway analysis, we can conclude that MLGCN-Driver has excellent performance in predicting driver genes. These genes are not only directly associated with the development of cancer but also have an impact on the progression of cancer in biological process, molecular function, and cellular component.

Conclusion and discussion

In this work, we propose a novel method, MLGCN-Driver, to identify cancer driver genes. MLGCN-Driver utilizes both biological features and network topological features extracted by node2vec to assist the model in making predictions. Both biological and network topological features are fed into two separate multi-layer GCN modules for node feature learning. During the feature learning process using multi-layer GCN, initial residual connections and identity mappings are employed to address the issue of driver gene features being smoothed by surrounding non-driver gene features and the over-smoothing problem that arises from excessively deep GCN layers. Finally, node feature representations are obtained by weighing the learned low-dimensional node representations and they are fed into the predictor for prediction. We conducted experiments on pan-cancer datasets and cancer type-specific datasets, and the results demonstrate that

the proposed MLGCN-Driver method possesses certain advantages. MLGCN-Driver achieved the optimal performance on both types of datasets. The ablation experiments demonstrate that the model incorporating initial residual connections, identity mappings, and network topological features extracted by node2vec outperformed all its variant models. Among these variant models, MLGCN-Driver achieved the highest AUROC and AUPR values.

MLGCN-Driver utilizes a combination of biological features and network topological features to identify cancer driver genes. However, relying solely on these features is insufficient to describe the complex molecular mechanisms of cancer driver genes in detail. One potential direction for future research is to extract features that can better capture the molecular mechanisms of cancer driver genes. Additionally, due to the limited number of known cancer driver genes and the lack of reliable driver gene databases, the issue of class imbalance needs to be considered and tackled by implementing corresponding strategies to mitigate the negative impact on cancer driver gene prediction. Finally, since multi-layer GCN has been employed for node feature learning, future work could explore how to integrate attention mechanisms to further enhance the model's performance.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12859-025-06260-8.

Supplementary Material 1

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Not applicable.

Author contributions
P.J.W. provided guidance, revised the manuscript, and supervised the research project. J.X.Z. conducted the literature review, developed the model code, wrote the original manuscript, and revised the manuscript. R.F.C., Y.D., Z.Y., and C.H.Z. provided guidance and assisted in revising the manuscript.

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Data availability
The code and data are available at https://github.com/ZhouJxz0516/MLGCN-Driver. The data is obtained from Zhang et al.'s paper [36].

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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